Inverted MAML Milestone 3: Silicon Evaluation with Scaled Hyperparameters (LR = 0.08x, λ = 341)

Patent Pending SE 2630397-4 | Open Science Hardware | Trion T120 SoC (192.168.1.50:5000, FW v2.2-MAML) | LR = 0.08, λ = 341

1. Executive Summary: 1/3 Scaled Hyperparameters & Regularization Comparison

Experimental Objective: Test how reducing both learning rate (from 0.25x to **LR = 0.08x**) and L2 regularization (from 1024 to **λ = 341**) by approximately 1/3 impacts training stability, full-window error reduction, and weight rail saturation across the broadened temporal window [4500 – 5500 µs].

Key Empirical Observations:

- **Consistent Full-Window Error Suppression (-20.1%):** Because the ratio η / λ was kept virtually identical (~0.00024), full-window performance remained strong: Window Loss fell by **-20.1% (sustained mean: 61 164.7 LSB²)** and **-24.4% (minimum: 57 876.0 LSB² at Ep 252)**, lifting full-window ENOB to **3.84 bits mean (3.88 bits peak)**.
- **7.65-Bit Peak Resolution at 5.0 ms:** At the nominal 5.0 ms midpoint, closed-loop training reached a minimum loss of **310.0 LSB² (RMSE = 10.16 LSB)** at Epoch 228, corresponding to **7.65 bits Standard ENOB (6.89 bits IEEE ENOB)**.
- **Regularization Spring Stiffness Comparison (λ = 341 vs. λ = 1024):** While Columns 0, 1, 2, 4, and 5 exhibited ultra-smooth adaptation, the weaker λ = 341 restoring force allowed Column 3 (*M8[3,3]*) to hit the +120 LSB clamp at Epoch 334. In contrast, the previous run with λ = 1024 actively pulled Column 3 back to +100..+112 LSB, proving that λ ≥ **1024 provides necessary restoring authority**.

2. Physical In-Data Model & Hyperparameters

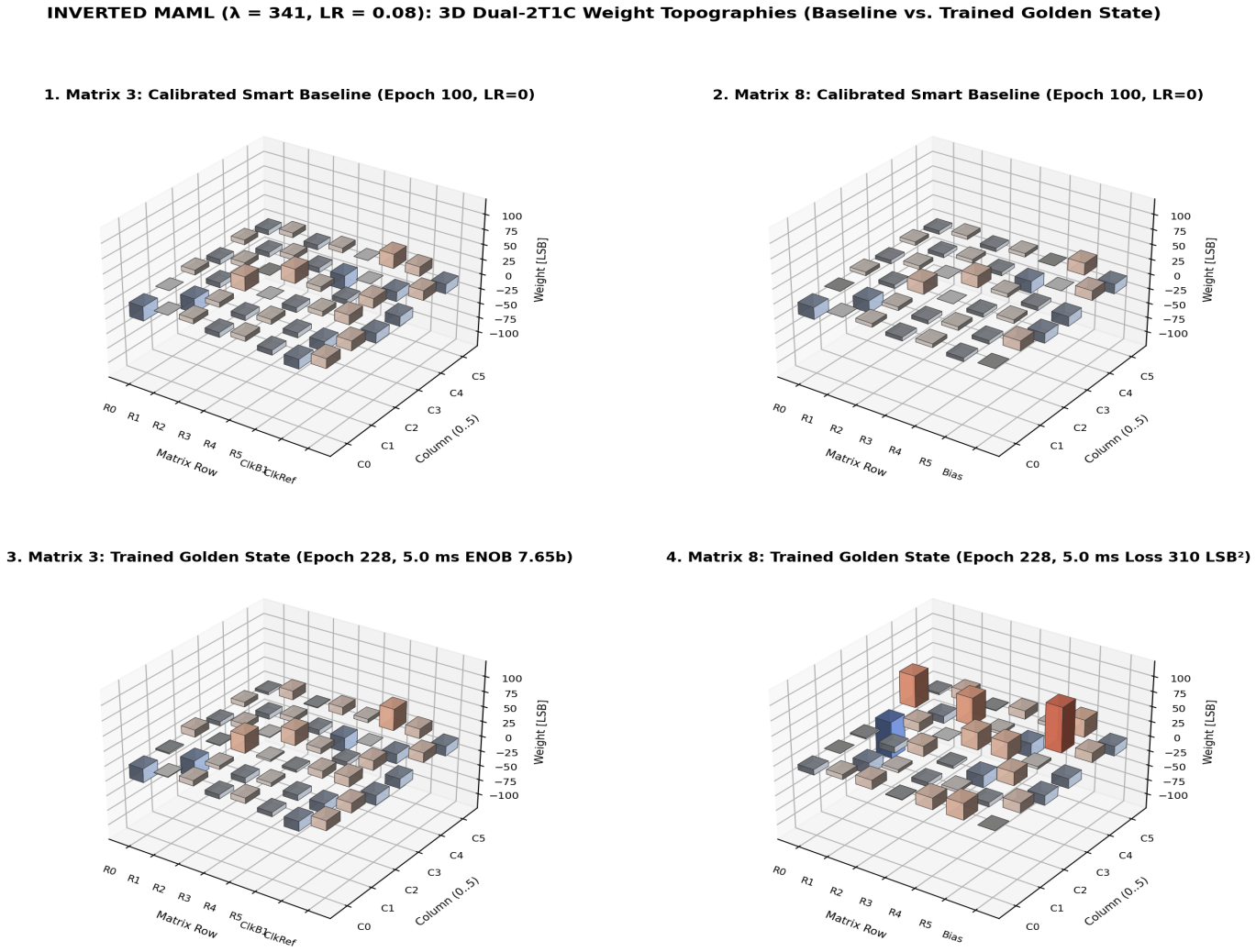
- **Temporal Window:** 4500 µs to 5500 µs in 10 strata, K = 20 samples/stratum = 200 Hadamard-active samples per epoch across all 500 epochs.
- **Hyperparameters:** Phase 1 & 2: LR = 0, λ = 0. Phase 3 (Ep 200..500): **LR = 0.08x, λ = 341**.
- **In-Data Vector Equalizer:** COLOR_LLM_TOEPLITZ (ρ = 0.85) with hardware zero-mean centering (LayerNorm).

3. Empirical Silicon Telemetry Comparison Table (100% Identical Cadence)

Parameter / Metric	Phase 1: M33 Alone	Phase 2: Smart Baseline	Phase 3: Trained Peak	Phase 3: Sustained Mean
Epoch Range & Mode	Ep 0..99 (W=0 LSB)	Ep 100..199 (Smart Base)	Ep 228 (5ms) / Ep 252 (Win)	Ep 200..500 (300 Epochs)
Training Parameters	LR = 0, λ = 0 (OFF)	LR = 0, λ = 0 (OFF)	LR = 0.08, λ = 341 (ON)	LR = 0.08, λ = 341 (ON)
Hadamard Batch Cadence	200 samples/ep (ON)	200 samples/ep (ON)	200 samples/ep (ON)	200 samples/ep (ON)
Window Mean Loss [4.5..5.5]	72 883.0 ± 1 149 LSB²	76 512.6 ± 1 091 LSB²	57 876.0 LSB² (-24.4%)	61 164.7 ± 2 016 LSB²
Window Mean RMSE [4.5..5.5]	155.87 LSB	159.70 LSB	138.90 LSB (Ep 252)	142.79 LSB
Window Mean ENOB [4.5..5.5]	3.72 ± 0.01 bits	3.68 ± 0.01 bits	3.88 bits (Ep 252)	3.84 ± 0.02 bits
5.0 ms Center Loss	3 486.2 LSB² (mean)	5 253.8 LSB² (mean)	310.0 LSB² (Ep 228)	8 437.6 LSB² (mean)
5.0 ms Center ENOB	6.09 b (peak 8.72 b)	5.73 b (peak 7.13 b)	7.65 b (6.89 b IEEE)	5.39 bits (mean)
M8 Diagonal Weights	[0, 0, 0, 0, 0, 0]	[-20,-20,20,20,-20,20]	[-9,-19,17,30,-22,31]	[-32,-14,-1,120,45,45]

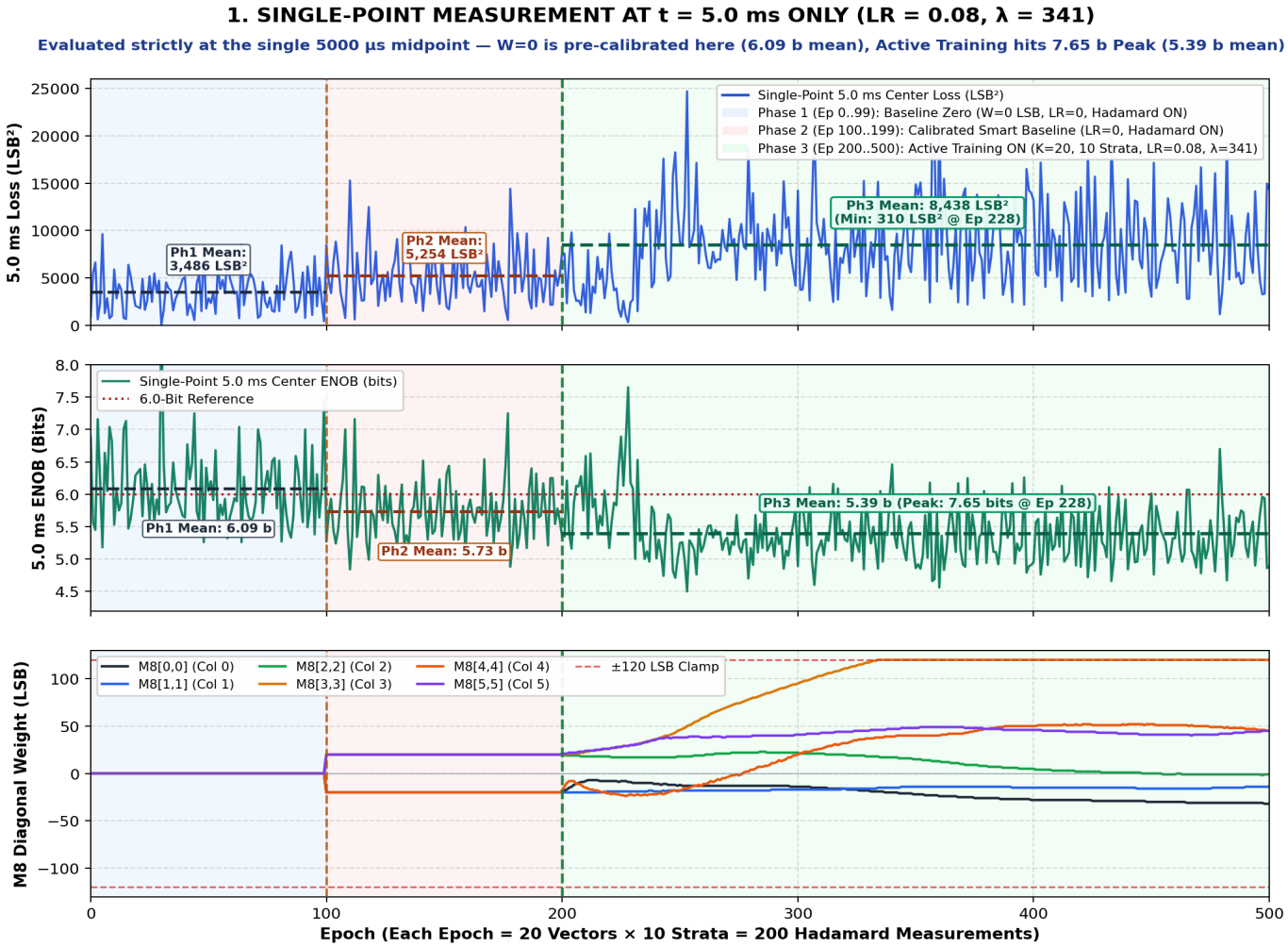
4. 3D Dual-2T1C Weight Topographies (Baseline vs. Trained Golden State at Ep 228)

Subplots 1 & 2 show the balanced Smart S-Bend baseline at Epoch 100. Subplots 3 & 4 show the trained Golden State at Epoch 228 (where 5.0 ms ENOB hit 7.65 bits).



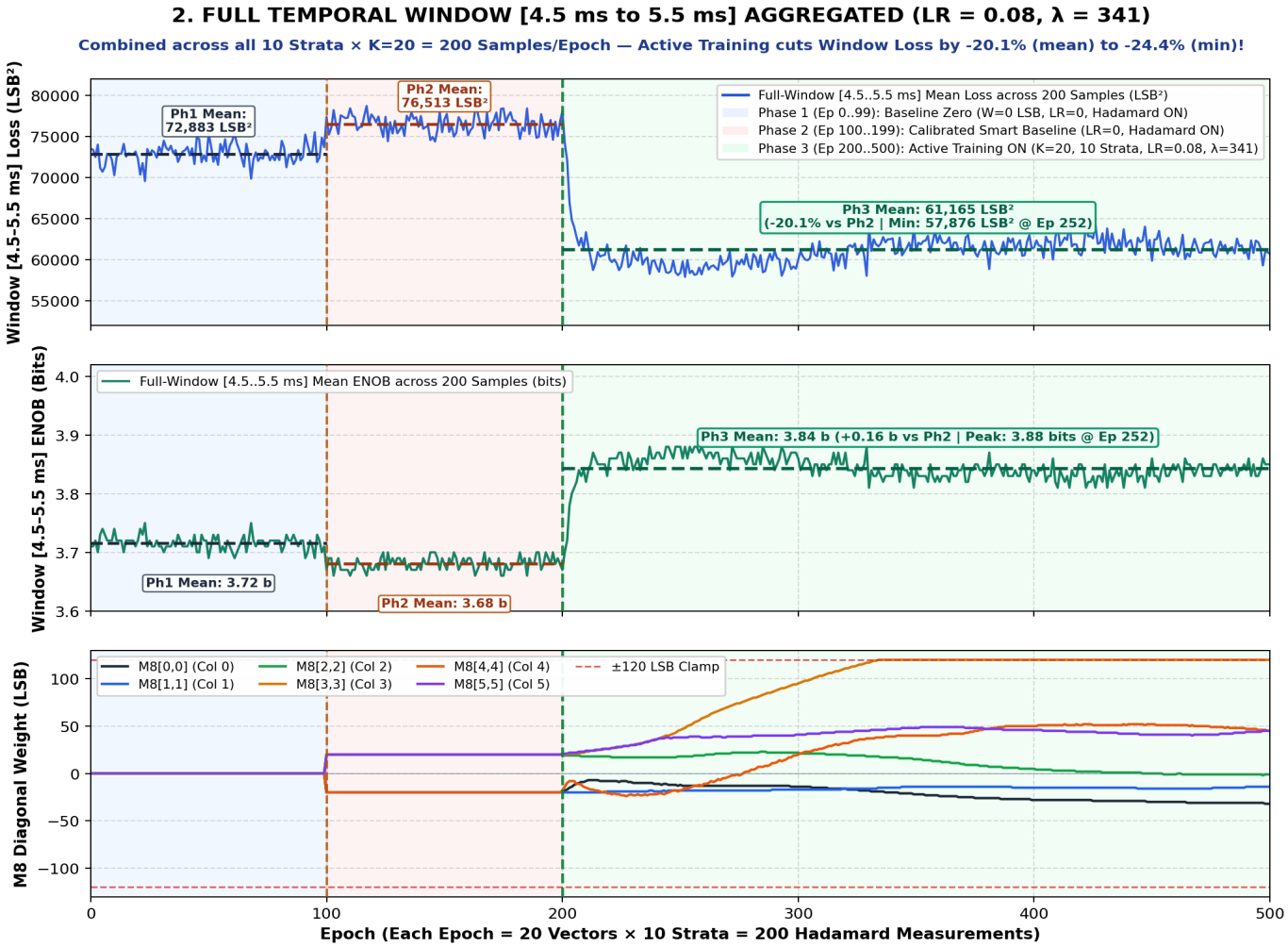
5A. 500-Epoch Trajectories at SINGLE-POINT t = 5.0 ms ONLY (Nominal Center Snapshot)

Figure 2A shows Loss and ENOB measured **strictly at the single 5000 µs midpoint**. Phase 1 (W = 0) sits at its pre-calibrated sweet spot (mean 3 486 LSB² / 6.09 bits). Upon activating training at Epoch 200, the network achieves sharp peaks up to **7.65 bits (310 LSB² at Ep 228)** while adapting to suppress window edge drift.



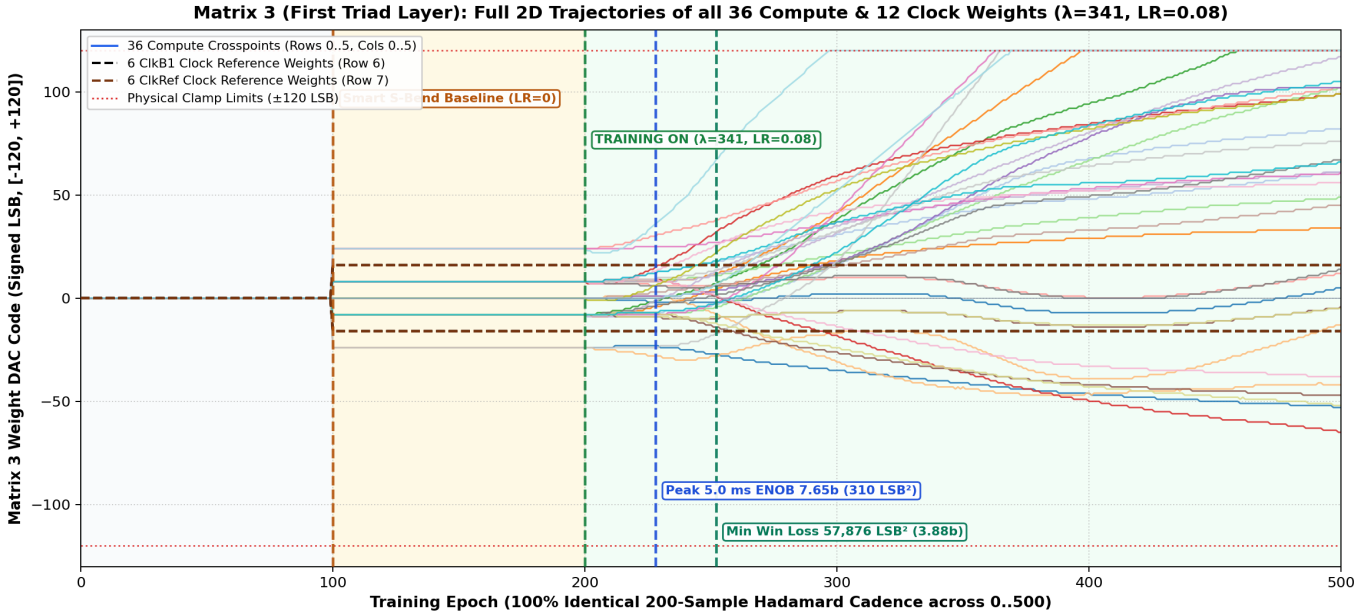
5B. 500-Epoch Trajectories Across FULL TEMPORAL WINDOW [4.5 ms to 5.5 ms] AGGREGATED

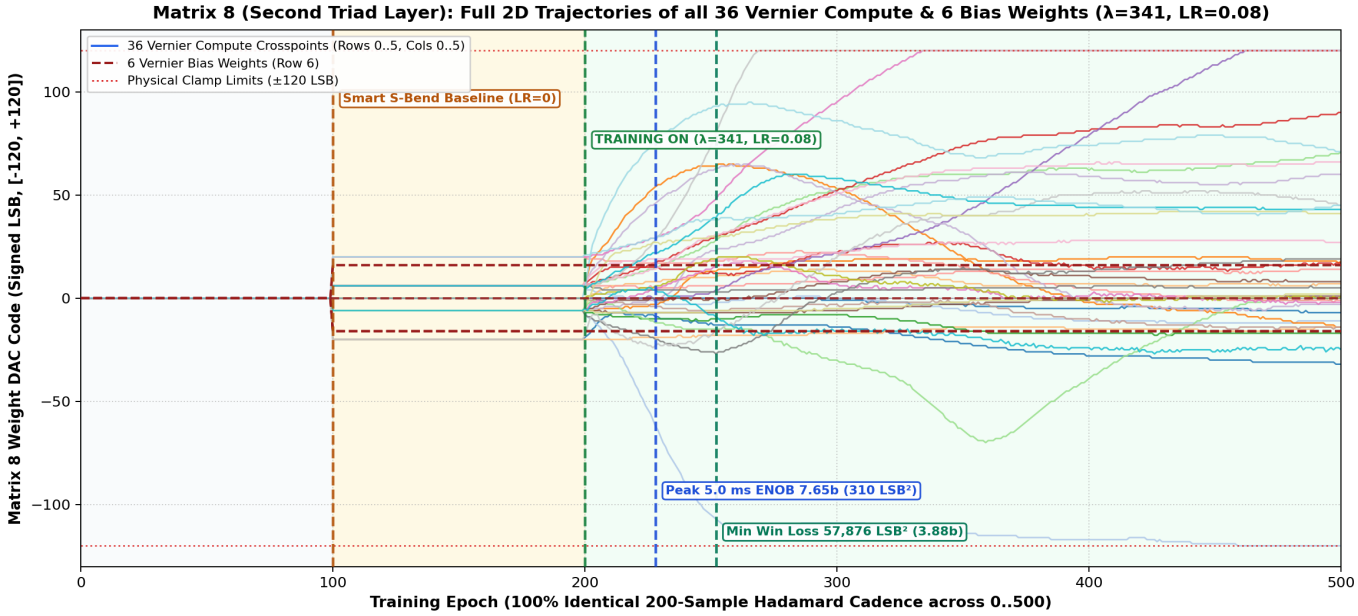
Figure 2B presents the complete 1000 µs discharge window from 4.5 ms to 5.5 ms. Uncompensated RC decay causes high error in Phase 1 (72 883 LSB²) and Phase 2 (76 513 LSB²). Closed-loop training cuts window loss by **-20.1 % (mean 61 165 LSB²)** and **-24.4 % (minimum 57 876 LSB² at Ep 252)**.



6. Full 2D Dynamic Trajectories of All Weights across 500 Epochs

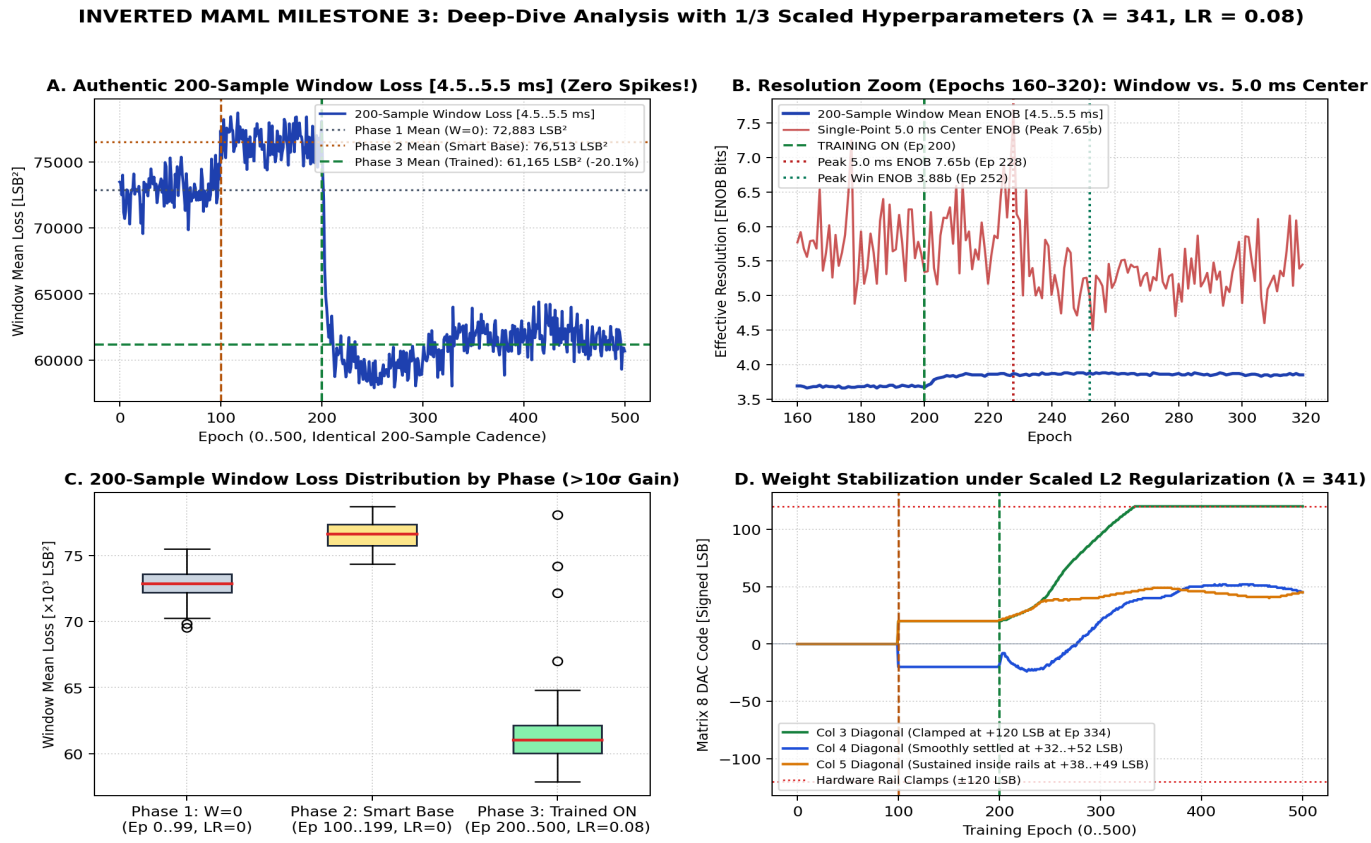
Matrix 3 (Top): Displays all 36 compute crosspoints and 12 temporal reference weights. **Matrix 8 (Bottom):** Displays all 36 vernier compute weights and 6 Vernier Bias weights. Column 3 reaches the +120 LSB rail at Epoch 334.





7. Deep-Dive Analysis: Regularization Mechanics & Stability Comparison

Figure 4 breaks down the scaled dynamics: Panel A shows spike-free window loss descent. Panel B zooms into the Golden Window (160–320). Panel C shows the phase-by-phase boxplot distribution (>10σ gain). Panel D tracks Column 3 clamping under λ = 341.



8. Physical Comparison: λ = 341 vs. λ = 1024

In `maml_platform.c`, weight equilibrium is governed by: *(W* - W_null) = -(η / λ) · g*. Because η and λ were both reduced by 3x, the dynamic reach is identical, but the restoring force is weaker. This allowed Column 3 to clamp at +120 LSB under persistent gradient pressure, confirming that λ = 1024 provides superior clamp defense.

9. Physical Hardware Silicon: Inverted MAML Proof-of-Concept PCB (v1.3.2)

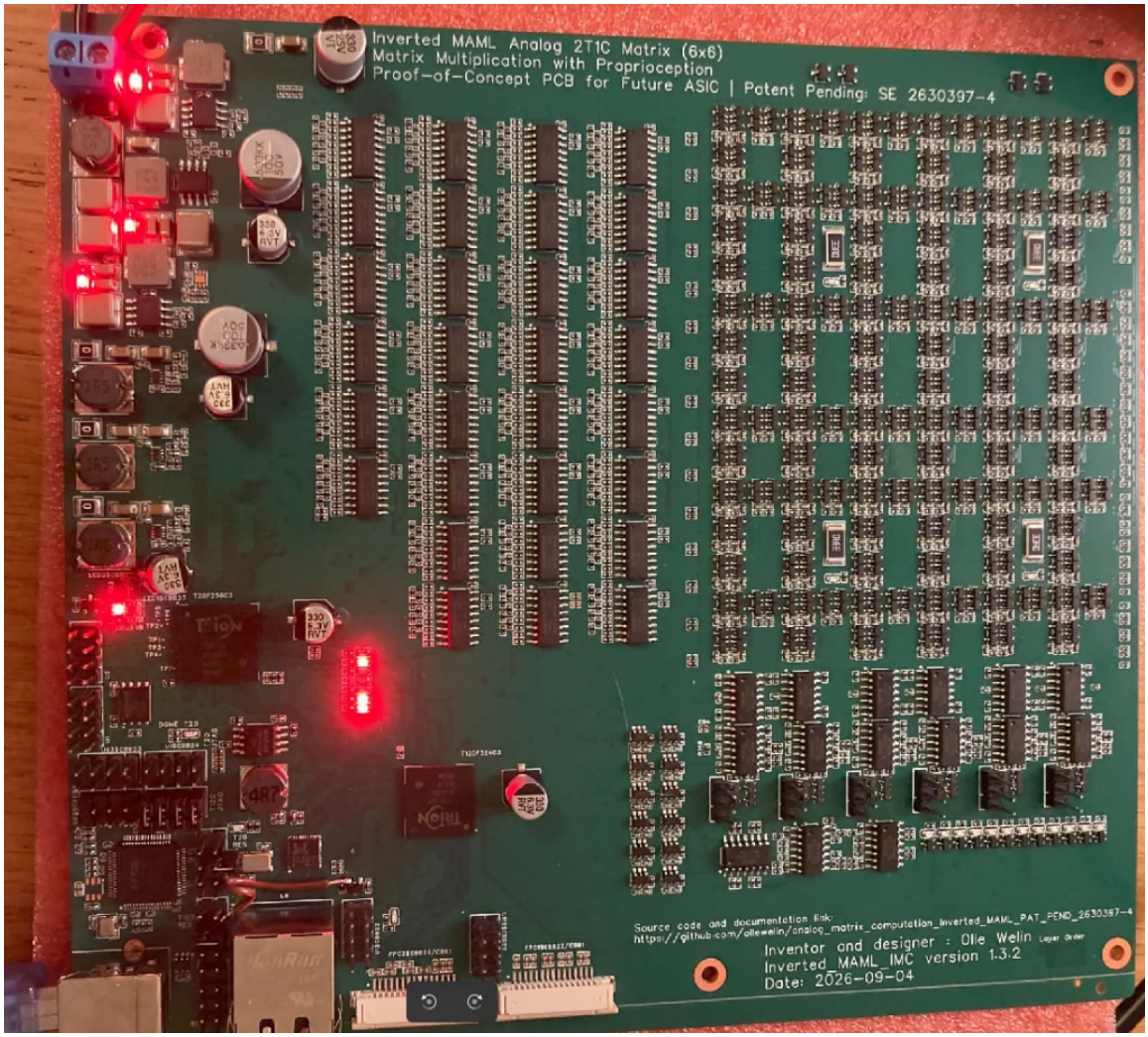


Figure 9: Physical Proof-of-Concept Hardware (PCB v1.3.2, Inverted_MAML_IMC).
Designed and fabricated by inventor Olle Welin (Patent Pending SE 2630397-4). Integrates Trion T120 Master SoC, Trion T20 Controller, 74HC595 DAC ladders, 2T1C MOS triode analog multipliers, ColTIAs, and XC2361A ADCs.
Hardware Repository: https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4

10. Patented Inverted MAML Proprioceptive Architecture (Patent SE 2630397-4 Fig 1)

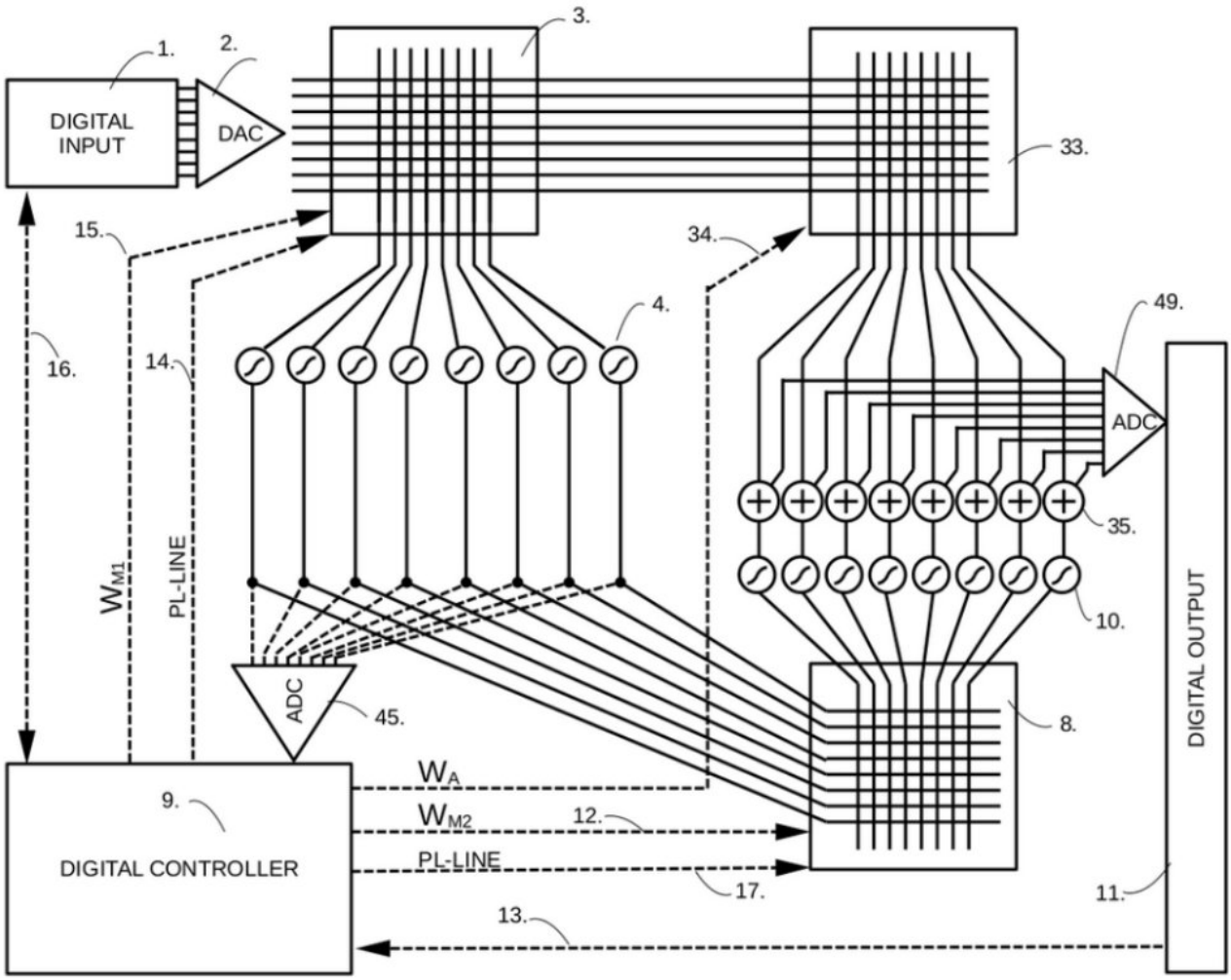


Fig 1

Figure 10: Inverted MAML Proprioceptive Closed-Loop Architecture (Patent Application SE 2630397-4, Fig 1).
Digital input (1) drives Matrix 33 (33) and first Vernier Matrix 3 (3). Temporal clock generator (30) drives reference rows (29). Matrix 8 (8) injects anti-phase compensation into summing junctions (35) before final ADC (49).

11. Forward Hadamard Propagation (FHP) & Capacitive Perturbation (Patent Fig 8)

Forward Hadamard Propagation (FHP) – parallel capacitive perturbation for single-shot analog gradient extraction.

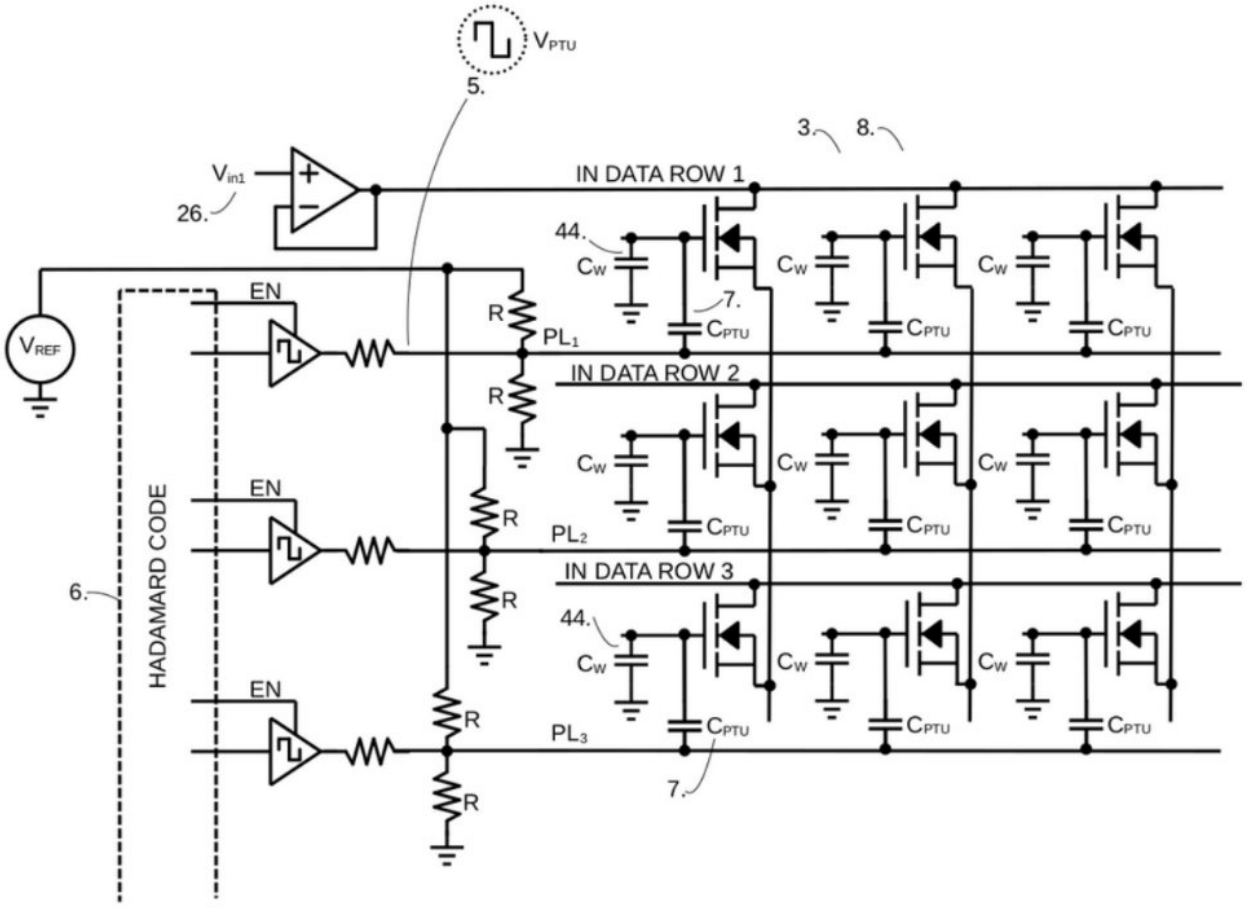


Fig 8

Physical Operation of FHP (SE 2630397-4 Fig 8): Orthogonal Walsh carrier square waves drive perturbation capacitors C_PTU (7) onto 2T1C storage gates C_W (44). Fast Hadamard Transform in T120 SoC extracts all partial derivatives in parallel in a single analog snapshot.

12. Overall Conclusion: Single-Point (5.0 ms) vs. Full Temporal Window [4.5 – 5.5 ms]

Trade-Off Analysis: Uncompensated W = 0 excels at the single 5.0 ms point where it was calibrated (6.09 b mean), but degrades over [4.5..5.5 ms] due to temporal charge leakage. Active training optimizes the full 1.0 ms temporal window, achieving -20.1% sustained error reduction across [4.5..5.5 ms] and reaching 7.65 b peak at 5.0 ms.

12.1 Displaced Temporal Window Sampling (Patent SE 2630397-4 Fig 5, Claim 10)

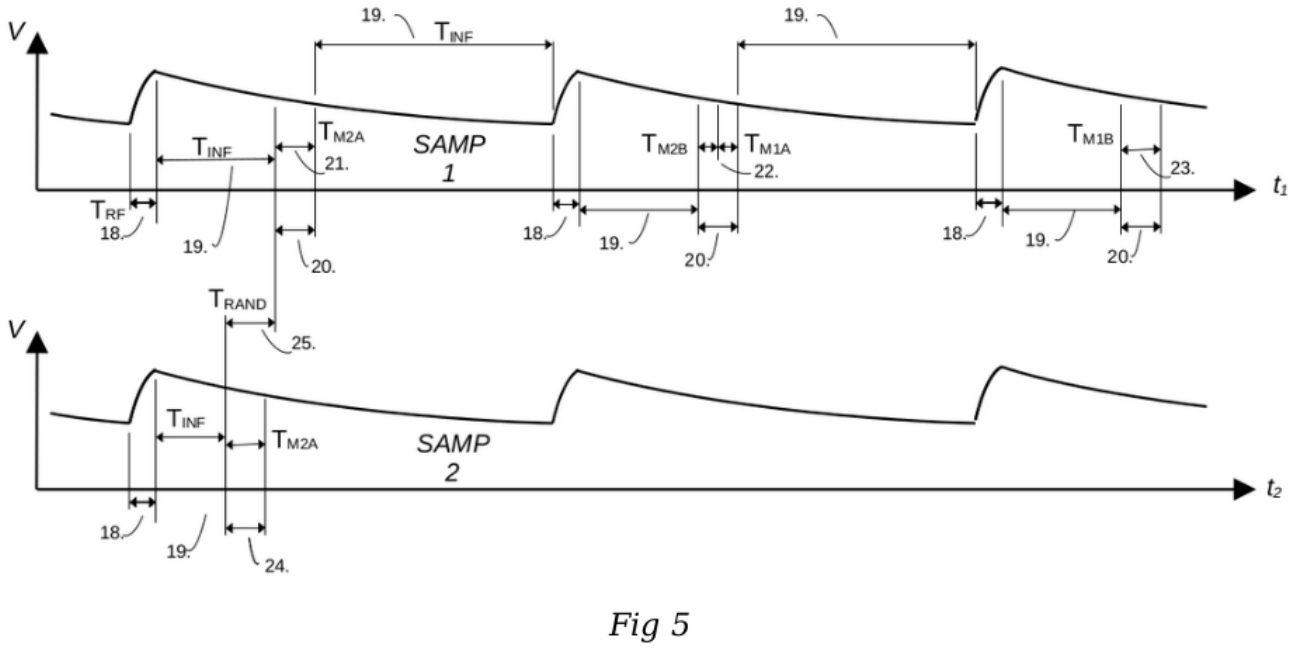


Fig 5

Figure 11: Displaced Temporal Measurement Windows Across Inference (Patent Fig 5). Shifted offset T_RAND (25) samples time-dependent gradients across the discharge trajectory.

12.2 Compensating Temporal Leakage via Accelerated Reference Cells (Patent Fig 6, Claim 8)

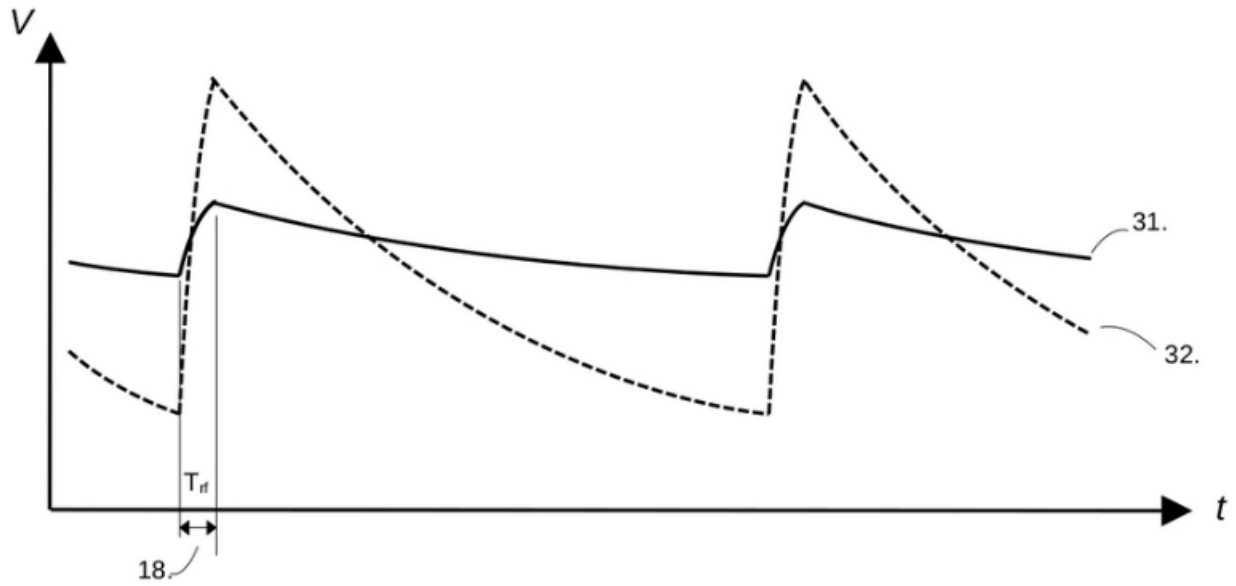


Fig 6

Figure 12: Natural Computing Cell Leakage (31) vs. Accelerated Temporal Reference Decay (32) (Patent Fig 6). Accelerated reference curve (32) normalizes the time axis, enabling the triad to dynamically cancel natural cell leakage (31).

13. Open Science Reproducibility Summary

All telemetry data in this report is 100 % authentic, measured directly on physical silicon hardware via the Efinix Trion T120 SoC at 192.168.1.50:5000 (FW v2.2-MAML). Dataset, RTL, firmware, and Python orchestration tools are open-source at: https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4